

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Systems for Data Science, Optimization in CS, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, SQL, C++, Java, Bash, HTML/CSS

LLM & Agent Orchestration

LLM agents, multi-agent systems, tool use, function calling, agent orchestration, durable workflows, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama, MCP

Full-Stack & Real-Time UIs

React, Electron, WebSockets, REST APIs, async services, data-intensive state management, FastAPI, Node.js, Express

Data & Storage

MongoDB, PostgreSQL, Redis, SQLite, Apache Spark / PySpark, Databricks, sentence-transformers, RAG

Cloud, Infra & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD, observability / instrumentation

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected and shipped a **data-intensive full-stack tool** (React, Node.js, MongoDB) used daily by the AI team to ingest, label, and surface high-volume training data through reviewer dashboards with complex state management; cut manual operations by **100+ hours/month** and lifted throughput **60%**.
- Owned multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; hardened auth, rate limiting, and remediated **20+ VAPT findings** (contributed to **\$340K+** in revenue).
- Served as **primary technical point-of-contact** for the eTriage backend; shipped audit logging, monitored production logs during feature releases and client integrations, triaged tenant-specific incidents, and ran knowledge transfer for new developers on the UK Clinical team.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped **LLM-powered context-aware replies** into a customer-facing chat product, iterating on prompt design and response-quality heuristics from real usage; designed **20+ production RESTful APIs** in Node.js + Fastify over PostgreSQL with clear contracts and **JS→TS migration** across multiple repositories.

Projects

cf-ai-research-scout - Real-Time AI Agent with Durable Orchestration

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an **AI agent** end-to-end with a TypeScript frontend that fetches web sources, generates multi-step research digests, and **streams agent-generated insights to the client over WebSocket in real time**, surfacing live progress as the agent works.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step agent tasks survive process restarts and tool failures, with agent state and conversation history persisted in D1 without an external database.

AgenticSearch - Multi-Stage Agent with Tool Integrations and Trust UI

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic service with **tool integrations** (Brave Search, deterministic parsers, LLM fallback) and **agent orchestration** that converts open-ended queries into structured entity tables with cell-level provenance.
- Hardened the service with **150+ automated tests**, **provider routing/fallback across OpenAI and Groq**, pipeline instrumentation, and a reviewer-facing trust UI that surfaces agent decisions, source URLs, and ranking transparency as a product experience.

Sonare - Full-Stack Real-Time Audio/Video Pipeline

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Took a **cross-platform desktop product** from zero to one under hackathon time pressure: React/Electron frontend, FastAPI backend, on-device pipeline integrating MediaPipe hand tracking, whisper.cpp ASR, and local TTS, with async APIs and queues keeping the UI responsive over high-volume video and audio streams.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.